# 2-1 Cardiac arrest v.1

The probable cause is one or more of: something related to surgery or anaesthesia; the patient's underlying medical condition; the reason for surgery; equipment failure. The first priority is to start chest compressions, then get help, then find and treat the cause using the guideline.

## START

**1 IMMEDIATE ACTION**

- **Declare "cardiac arrest"** to the theatre team AND note time.
- Delegate one person (minimum) to chest compressions 100 $min^{-1}$, depth 5 cm.
- Call for help: nearby theatres / emergency bell / senior on-call / dial emergency number.
- Call for cardiac arrest trolley.
- As soon as possible, delegate task of evaluating potential causes (Box A).

**2 Adequate oxygen delivery**

- Increase fresh gas flow, give 100% oxygen AND check measured $F_iO_2$.
- Turn off anaesthetic (inhalational or intravenous).
- Check breathing system valves working and system connections intact.
- Rapidly confirm ventilator bellows moving or provide manual ventilation.

**3 Airway**

- Check position of airway device and listen for noise (including larynx and stomach).
- Confirm airway device is patent (consider passing suction catheter).
- **If expired $CO_2$ is absent, presume oesophageal intubation until absolutely excluded.**

**4 Breathing**

- Check chest symmetry, rate, breath sounds, $SpO_2$, measured expired volume, $ETCO_2$.
- Evaluate the airway pressure using reservoir bag and APL valve.

**5 Circulation**

- Check rate and adequacy of chest compressions (visual and $ETCO_2$).
- Encourage rotation of personnel performing compressions.
- If i.v. access fails or impossible use intraosseous (IO) route.
- Check ECG rhythm for no more than 5 seconds.
- Perform FEEL Echo for no more than 10 seconds.
- Follow Resuscitation Council (UK) and ERC Guidelines.
- See Boxes B and C for reminders about drugs and defibrillation.

**6 Systematically evaluate potential underlying problems and act accordingly** (Box A).

**7** If there is return of spontaneous circulation, re-establish anaesthesia.

### Box A: POTENTIAL CAUSES

***4 H's, 4 T's:***

Hypoxia **(→ 2-2)**
Hypovolaemia
Hypo/hyperkalaemia
Hypothermia

Tamponade **(→ 3-9)**
Thrombosis **(→3-5)**
Toxins
Tension pneumothorax

**Specific peri-operative problems:**

Vagal tone
Drug error
Local anaesthetic toxicity **(→ 3-10)**
Acidosis
Anaphylaxis **(→ 3-1)**
Embolism, gas/fat/amniotic **(→ 3-5)**
Massive blood loss **(→ 3-2)**

### Box B: DRUGS FOR PERI-OPERATIVE CARDIAC ARREST

Fluid bolus 20 $ml.kg^{-1}$ (adult 500 ml).
Adrenaline 10 $\mu g.kg^{-1}$ (adult 1000 µg – may be given in increments).
Atropine 10 $\mu g.kg^{-1}$ (adult 0.5-1 mg) if vagal tone likely cause.
Amiodarone 5 $mg.kg^{-1}$ (adult 300 mg) after 3rd shock.
Magnesium 50 $mg.kg^{-1}$ (adult 2 g) for polymorphic VT/hypomagnesaemia.
Calcium chloride 10% 0.2 $ml.kg^{-1}$ (adult 10 ml) for magnesium overdose, hypocalcaemia or hyperkalaemia.
Thrombolysis for suspected massive pulmonary embolus.

### BOX C: DEFIBRILLATION

Continue compressions while charging: Biphasic 4 $J.kg^{-1}$ (adult 150-200 J)
DO NOT check pulse after defibrillation.
Use 3 stacked shocks in cardiac catheterisation lab.

### BOX D: DON'T FORGET!

- **Use waveform capnography. No expired $CO_2$ = lungs not being ventilated (assume and exclude oesophageal intubation). Very rarely, absent/minimal expired $CO_2$ = CPR not occurring OR pulmonary circulation disconnected from systemic (e.g. in major trauma).** Sudden increase in $ETCO_2$ usually signals return of spontaneous circulation.
- Optimise position for chest compressions (use overhead for bariatric patients).
- Uterine displacement in pregnant patients.
- Ventilator can free up hands but remember to set to volume control. Minimise intrathoracic pressure: avoid excessive tidal volume and hyperventilation.

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